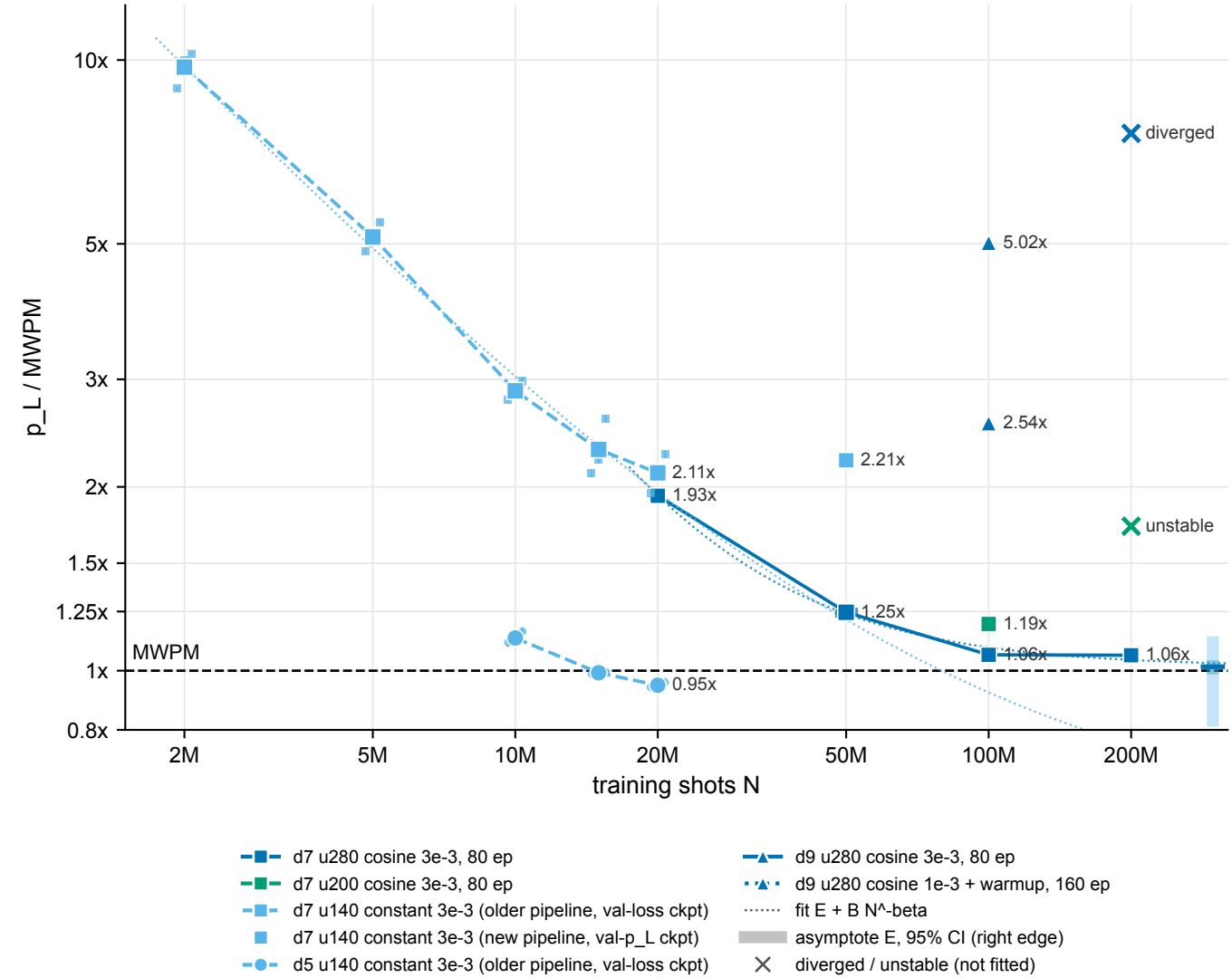
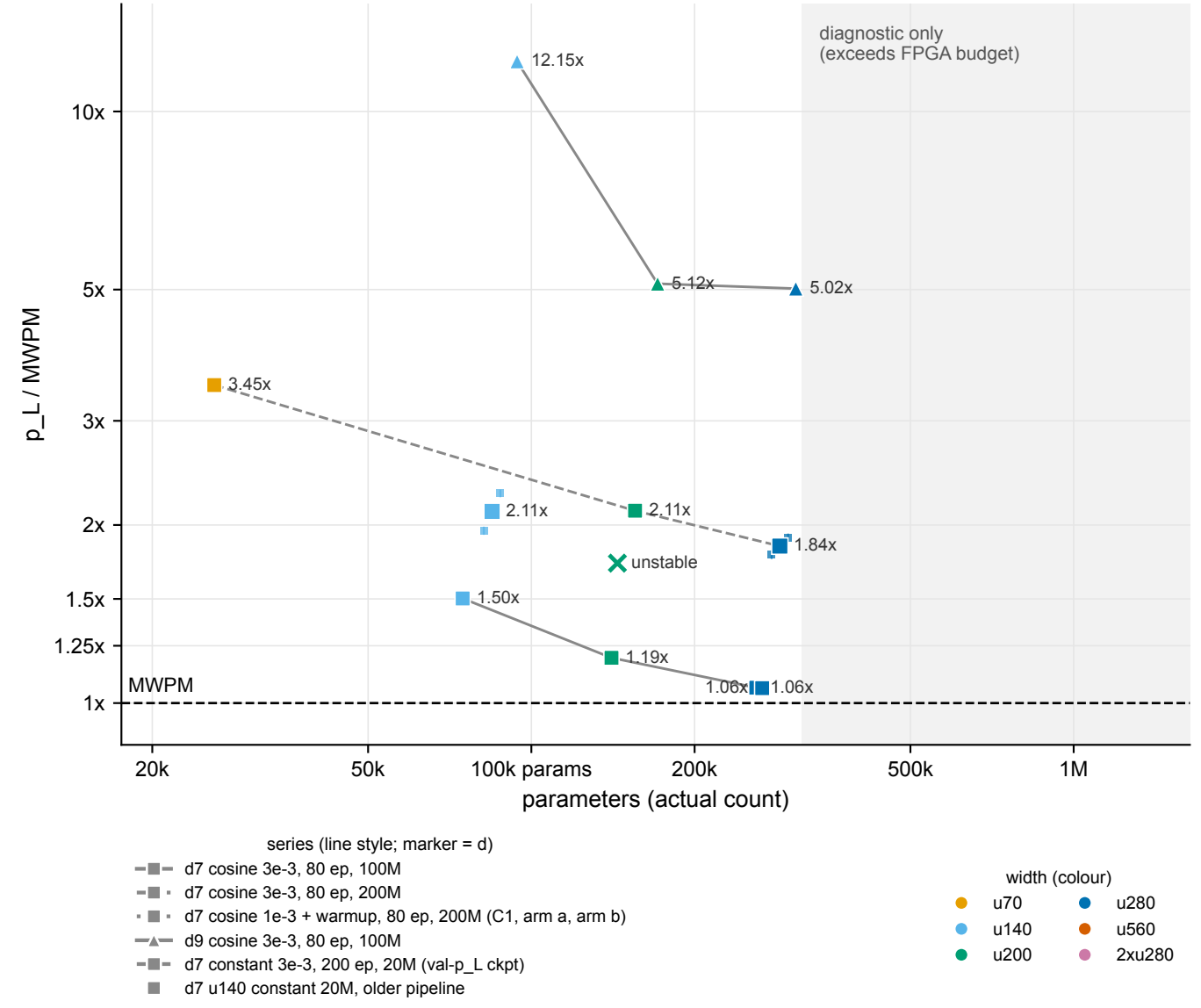
d7 u280: 1.93x MWPM at 20M, 1.06x at 100M, 1.06x at 200M; fitted floor 1.01x



series (fitted)	E (x MWPM) [95% CI]	beta [95% CI]	runs / distinct N
d7 u280 cosine 3e-3, 80 ep	1.01 [0.99, 1.04]	1.54 [1.39, 1.71]	5 / 4
d7 u140 constant 3e-3 (older pipeline, val-loss ckpt)	0.54 [0.00, 1.14]	0.81 [0.69, 0.97]	14 / 5

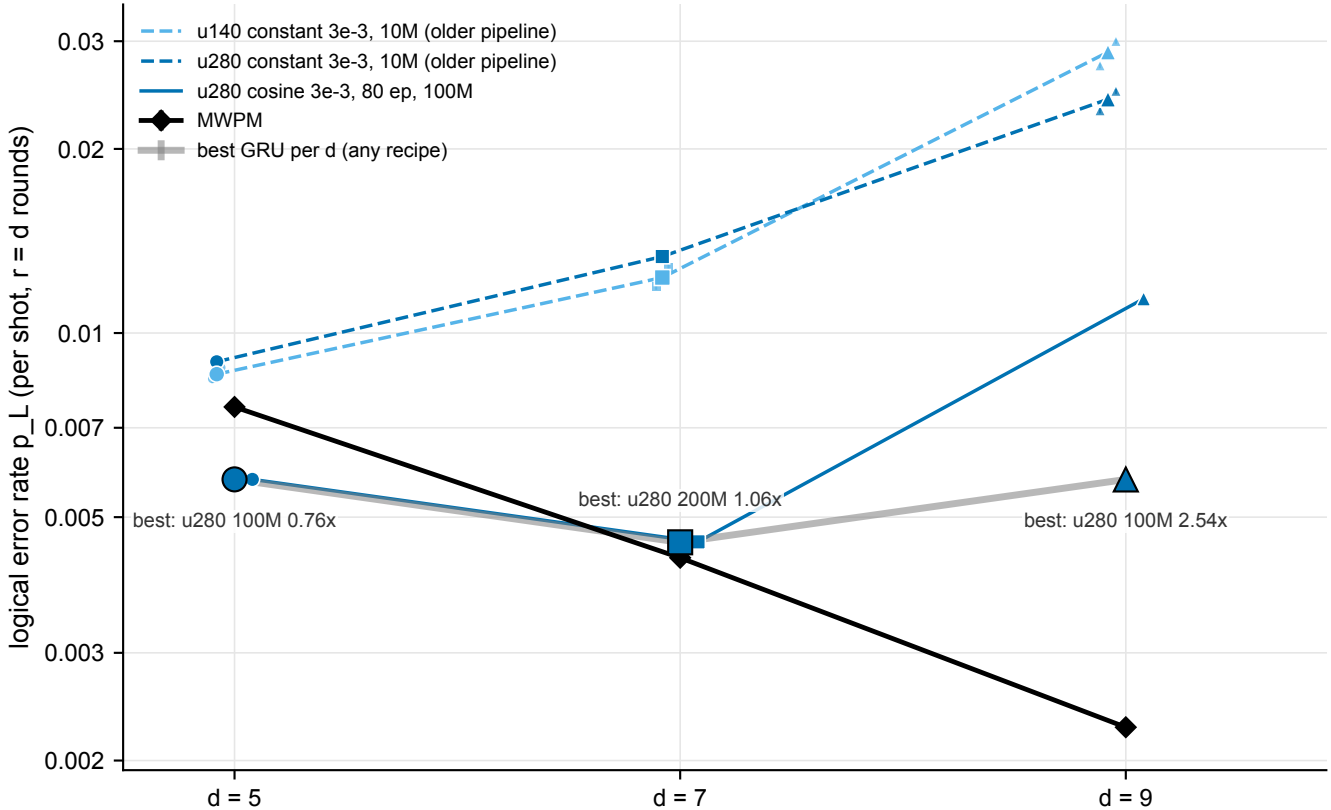
Each line = one recipe; only N changes. Solid: cosine 3e-3, 80 ep. Dashed: constant 3e-3, older pipeline. Dotted: d9 1e-3 + warmup, 160 ep. Fit E + B N^-beta where >= 4 N; bar at right = E, 95% CI. Pending: 9-5. Eval block: 1.8M shots per d, same MWPM for every run at a given d. Bars: Wilson 95%. Data: GRU_all_runs_2026-10-01.csv.

At fixed data and recipe, d7 error falls with width: u140 1.50x, u200 1.19x, u280 1.06x at 100M



Each line = one recipe at one data size. Shaded = above the u280 FPGA budget. Pending: arm a, arm b, C1.
Eval block: 1.8M shots per d, same MWPM for every run at a given d. Bars: Wilson 95%. Data: GRU_all_runs_2026-10-01.csv.

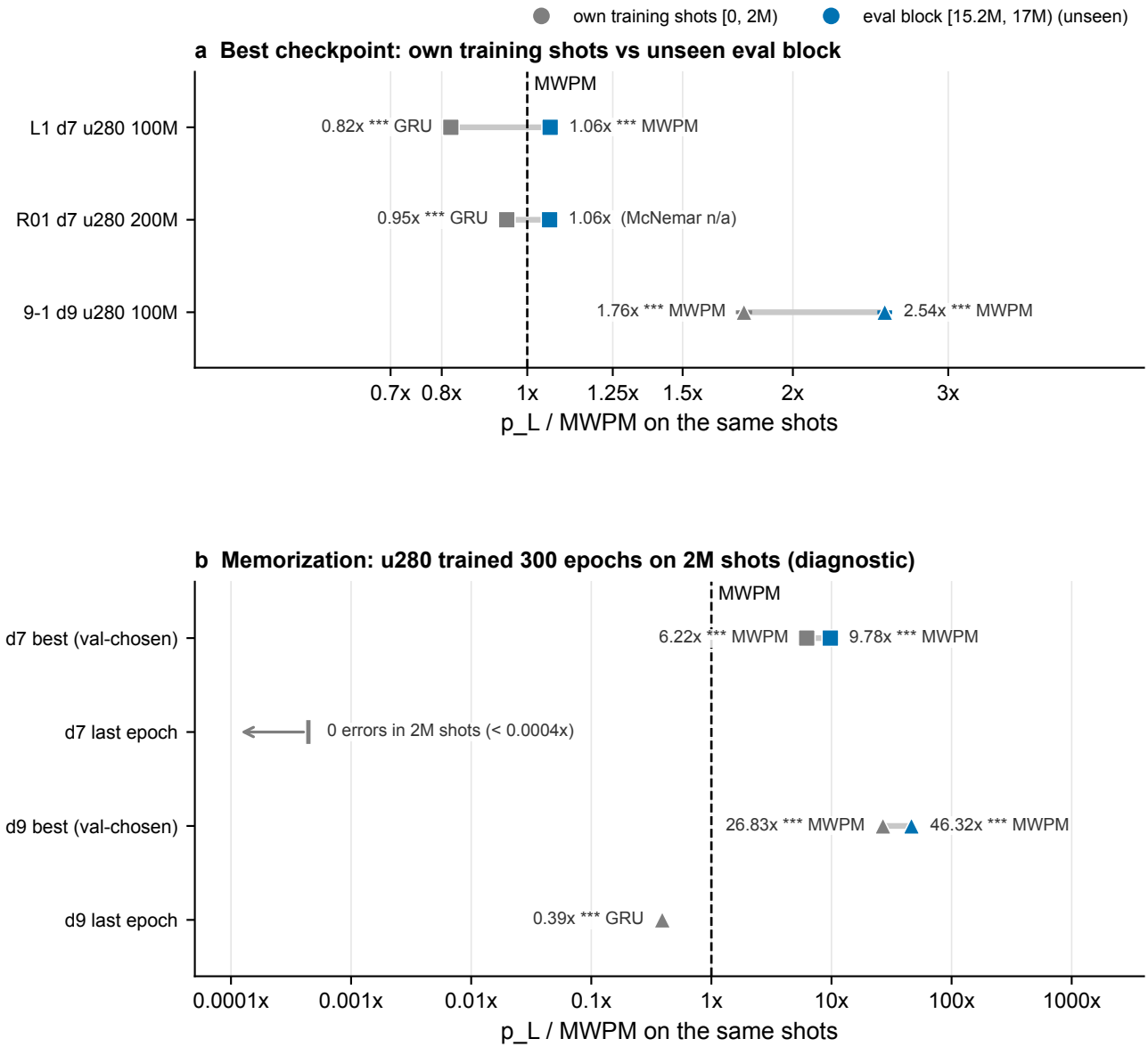
GRU Λ < MWPM Λ (5→7: 1.27 vs 1.76; 7→9: 0.79 vs 1.89); GRU Λ < 1 from d7 to d9



series	Lambda 5->7 [95% CI]	Lambda 7->9 [95% CI]
MWPM	1.76 [1.72, 1.81]	1.89 [1.82, 1.96]
best GRU per d	1.27 [1.23, 1.30]	0.79 [0.77, 0.81]
u140 constant 3e-3, 10M (older pipeline)	0.69 [0.67, 0.72]	0.43 [0.41, 0.45]
u280 constant 3e-3, 10M (older pipeline)	0.67 [0.66, 0.69]	0.55 [0.54, 0.58]
u280 cosine 3e-3, 80 ep, 100M	1.27 [1.23, 1.30]	0.40 [0.39, 0.41]

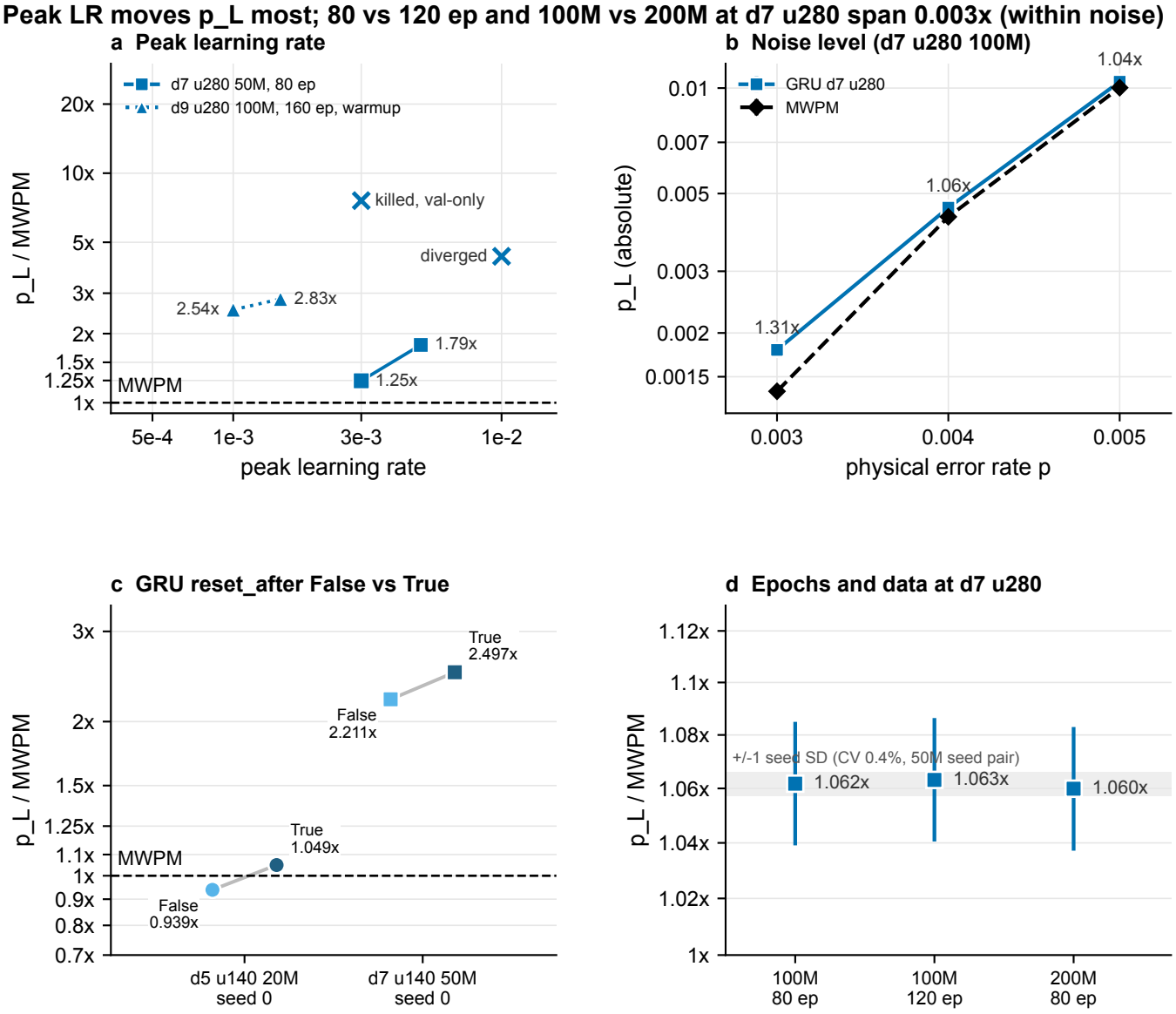
Lambda = p_L(d) / p_L(d+2), bootstrap 95% CI. Grey = best GRU at each d (any recipe); coloured lines use one recipe at every d. Eval block: 1.8M shots per d, same MWPM for every run at a given d. Bars: Wilson 95%. Data: GRU_all_runs_2026-10-01.csv.

Unseen p_L is 10-36% above own-training-shot p_L; 300 epochs on 2M shots memorizes

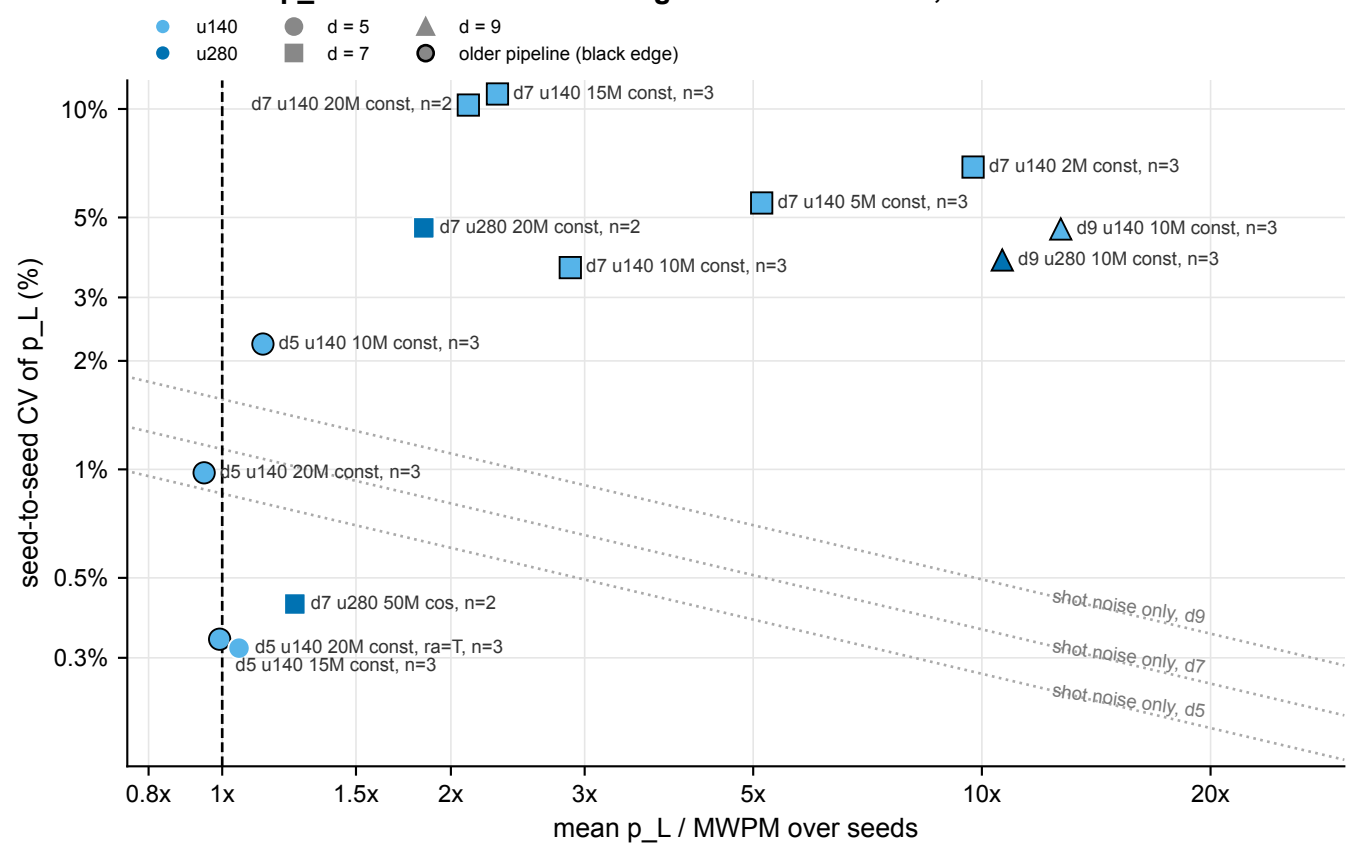


Last-epoch weights on unseen validation shots: d7 p_L 0.051; d9 p_L 0.132 (vs. 0.004-0.005 for the 100M runs)

(a) Best checkpoint on its own training shots vs the eval block; stars = McNemar vs MWPM (*** p<1e-3).
(b) Diagnostic: trained 300 ep on 2M shots only. Eval block: 1.8M shots per d, same MWPM for every run at a given d. Bars: Wilson 95%. Data: GRU_all_runs_2026-10-01.csv, figures/data/extra_scores.csv.



(a) Cosine, clip 0.64; x = diverged or killed. (b) Same run recipe at each p. (c) Matched seed and recipe only. (d) Band = seed-to-seed SD at 50M. Pending: 9-11. Eval block: 1.8M shots per d, same MWPM for every run at a given d. Bars: Wilson 95%. Data: GRU_all_runs_2026-10-01.csv.



One point per config with >= 2 seeds and one recipe. Dotted = CV expected from eval shot noise alone.
Eval block: 1.8M shots per d, same MWPM for every run at a given d. Bars: Wilson 95%. Data:
GRU_all_runs_2026-10-01.csv.